

Nazmul Huda Badhon

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[Personal Website](#) || [LinkedIn](#) || [Google Scholar](#)

Profile Summary

Computer Science and Engineering graduate with two years of research experience in deep learning and medical image analysis. Published eight peer-reviewed publications in top-tier journals. Experienced in neural network-based modeling and applied AI research. Seeking fully funded graduate research opportunities.

Academic Qualification

Bachelor of Science in Computer Science and Engineering

(Jan '22 - Dec '25)

Daffodil International University (DIU), Dhaka, Bangladesh

CGPA: 3.88 / 4.00 [Top 5% of cohort] || Thesis: Tabular data-based lightweight convolutional neural network.

Exchange Semester in Computer Engineering

(Sep '23 - Dec '23)

Dongseo University, Busan, South Korea

GPA: 3.94 / 4.50 [Received fully funded GKS-Global Korea Scholarship]

Research Interest

Designing computationally efficient deep learning architectures.

Medical Image Analysis and interdisciplinary AI research.

Research Experience

CSE AI Centre, Daffodil International University, Bangladesh

(Jan '26 - Present)

Graduate Researcher

Advised by: Associate Professor [Nazmun Nessa Moon](#) and Professor [Dr. Fernaz Narin Nur](#)

- Develop and organize research datasets for applied artificial intelligence projects.
- Design and evaluate deep learning experiments for medical imaging and computer vision tasks.
- Prepare research manuscripts and support collaborative publication activities.

Kyushu Institute of Technology, Kitakyushu, Japan

(Dec '25 - Mar '26)

Research Collaborator

- Analyzed real-world multimodal sensor data collected from a nursing care facility.
- Developed a Temporal Convolutional Network-based approach for sequence modeling.
- Classified indoor locations using BLE beacon signal data.

Health Informatics Research Lab, Daffodil International University, Bangladesh

(Jan '24 - Dec '25)

Researcher

- Developed deep learning models for medical image classification and segmentation tasks.
- Applied computer vision techniques to healthcare-related imaging datasets.
- Interpreted experimental results and contributed to research writing for publication.

Research Publications

Peer-Reviewed Journals:

1. **Nazmul Huda Badhon**, Sabit Ahmed Preanto, Abu Shahed Shah Md Nazmul Arefin, Kazi Jahid Hasan, and Md. Baharul Islam. [Temporal convolutional network based indoor location recognition from ble beacon signals in nursing care facility](#). *International Journal of Activity and Behavior Computing* (2026). Japan Challenge Paper
2. **Nazmul Huda Badhon***, Imrus Salehin*, Md. Tomal Ahmed Sajib, Md. Sakibul Hassan Rifat, S M Noman, Nazmun Nessa Moon. [TLCNN: tabular data-based lightweight convolutional neural network for electricity energy demand prediction](#). *Global Energy Interconnection* (2025), Elsevier. Q1
3. Md. Tomal Ahmed Sajib, **Nazmul Huda Badhon**, Imrus Salehin, Md. Sakibul Hassan Rifat, Faysal Ahmmed, Nazmun Nessa Moon. [A comparative deep learning methodology for plant insect image classification: assessment of CNN architectures and augmentation techniques](#). 103680, *MethodsX* (2025), Elsevier. Q2
4. Imrus Salehin, Md. Tomal Ahmed Sajib, **Nazmul Huda Badhon**, Md. Sakibul Hassan Rifat, Nazrul Amin, Nazmun Nessa Moon. [Systematic Literature Review of LLM-Large Language Model in Medical: Digital Health, Technology and Applications](#). 7.9: e70365, *Engineering Reports* (2025), Wiley. Q1
5. Imrus Salehin, **Nazmul Huda Badhon**, Md Tomal Ahmed Sajib, Mohd Saifuzzaman, and Nazmun Nessa Moon. [IQ-UltraRecon: Demodulated IQ ultrasound dataset of human hand and arm tissue for deep learning-based reconstruction](#). *Data in Brief* (2025), 112001, Elsevier. Q1
6. Imrus Salehin, **Nazmul Huda Badhon**, Md Tomal Ahmed Sajib, and Nazmun Nessa Moon. [DeepUS-ReconSeg: A multi-angle paired B-mode Ultrasound dataset for medical imaging reconstruction and segmentation](#). *Data in Brief* (2025), 112083, Elsevier. Q1
7. Imrus Salehin, Mahbubur Rahman Khan, Ummya Habiba, **Nazmul Huda Badhon**, Nazmun Nessa Moon. [BAU-Insectv2: An Agricultural Plant Insect Dataset for Deep Learning and Biomedical Image Analysis](#). *Data in Brief* (2024), 110083, Elsevier. Q1

Book Chapter:

8. Imrus Salehin, **Nazmul Huda Badhon**, Md. Tomal Ahmed Sajib, Md. Sakibul Hassan Rifat, Nazmun Nessa Moon. [Predicting Peak Energy Demand Using Machine Learning Techniques for Efficient Electricity Supply Management](#). In *Engineering Applications of AI for Demand Forecasting*, Vol. 1, p. 13. Taylor & Francis Group, 2026.

Manuscripts Under Review:

9. A. H. M. Saiful Islam*, **Nazmul Huda Badhon***, Fernaz Narin Nur*, Md. Tomal Ahmed Sajib, Nazmun Nessa Moon. Bridging Neuroscience and Education: A Scoping Review of EEG-Based Cognitive and Emotional States in Educational Settings. (Submitted to *International Journal of Educational Research Open*, Elsevier) Q1

* Joint co-first authors

Projects

Selected Research Projects

1. [Breast Cancer Classification using Hybrid Segmentation Deep Learning](#) 
 - Developed a hybrid U-Net and CNN model.
 - Evaluated model performance using accuracy and ROC-based metrics.
2. [Lung and Colon Cancer Histopathology Classification](#) 
 - Applied preprocessing and transfer learning techniques to histopathology images.
 - Evaluated classification performance using accuracy and F1-score.
3. [Vision Transformer–Based Medical Image Classification](#) 
 - Implemented a ViT-based classification pipeline for medical image analysis.
 - Achieved 87% test accuracy.

Additional Projects

4. [Lost and Found Web Application](#) 
 - Built a full-stack web application for reporting and retrieving lost items.
 - Used HTML, CSS, JavaScript, PHP, and MySQL with image upload functionality.

Technical Skills

Programming	: Python, Java, C
DL Techniques	: CNN, U-Net, Vision Transformers, Transfer Learning
Frameworks	: PyTorch, TensorFlow, Keras, Scikit-learn
Data Analysis	: NumPy, Pandas, Matplotlib, Seaborn
Research Tools	: GitHub, Google Colab, LaTeX

Language Proficiency

- Bangla: Native
- English: Professional working proficiency (Academic research, writing and presentation).

Academic Service

- Peer Reviewer, Digital Health, SAGE Publishing, 2025

Honors & Awards

- Research Award for Scholarly Publications, *Division of Research, DIU, 2024-2025*.
- Best Performer Award, *Computer Vision and Medical Image Analysis Bootcamp, HIRL, DIU, 2025*.
- Global Korea Scholarship awardee, *Government of the Republic of Korea, 2023*.
- Merit-based Scholarship, *Daffodil International University, 2022-2025*.
- Government General Board Scholarship, *Education Board Bangladesh, 2015*

Leadership and Extracurricular Activities

- **Team Leader** 2023
International Camp, Dongseo University, South Korea
- **Volunteer Intern** 2024
International Affairs, Daffodil International University
- **Founding General Secretary** (Mar '24 - July '24)
DIU Research Society

References

- **Nazmun Nessa Moon**
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